

## 20260930 Quiz

for (MATH200) Differential Equations

Jungwoo Kim

Department of Mathematics

Pohang University of Science and Technology

**Problem 2.7.3.** Consider the initial value problem

$$y' = 1 - t + 2y, \quad y(0) = 1.$$

- (a) Find approximate values of the solution at  $t = 0.1, 0.2, 0.3, 0.4$  using Euler's method with  $h = 0.1$ .
- (b) Repeat part (a) with  $h = 0.05$ . Compare the results with those found in part (a).
- (c) Repeat part (a) with  $h = 0.025$ . Compare the results with those found in parts (a) and (b).
- (d) Find the exact solution  $y = \phi(t)$  of the initial value problem and evaluate  $\phi(t)$  at  $t = 0.1, 0.2, 0.3, 0.4$ . Compare these values with the results of parts (a), (b), and (c).

*Solution.* (a)–(c) For Euler's method,

$$t_n = nh, \quad y_{n+1} = y_n + h(1 - t_n + 2y_n), \quad y_0 = 1.$$

Carrying out the recurrence gives

| $h$   | $y(0.1)$ | $y(0.2)$ | $y(0.3)$ | $y(0.4)$ |
|-------|----------|----------|----------|----------|
| 0.100 | 1.30000  | 1.65000  | 2.06000  | 2.54200  |
| 0.050 | 1.31250  | 1.68013  | 2.11445  | 2.62949  |
| 0.025 | 1.31938  | 1.69682  | 2.14482  | 2.67859  |

Thus, for each listed value of  $t$ , the Euler approximations increase as the step size decreases.

(d) The differential equation can be written as

$$y' - 2y = 1 - t.$$

An integrating factor is  $e^{-2t}$ , so

$$(e^{-2t}y)' = (1 - t)e^{-2t}.$$

Hence

$$y = ce^{2t} + \frac{t}{2} - \frac{1}{4}.$$

Using  $y(0) = 1$  gives  $c = 5/4$ , and therefore

$$\boxed{\phi(t) = \frac{5}{4}e^{2t} + \frac{t}{2} - \frac{1}{4}}.$$

The corresponding exact values are

|       | $\phi(0.1)$ | $\phi(0.2)$ | $\phi(0.3)$ | $\phi(0.4)$ |
|-------|-------------|-------------|-------------|-------------|
| Exact | 1.32675     | 1.71478     | 2.17765     | 2.73193     |

The Euler approximations move toward these exact values as  $h$  decreases.

□

**Problem 2.8.2.** Transform the initial value problem

$$\frac{dy}{dt} = 4 - y^3, \quad y(-1) = 2,$$

into an equivalent problem with the initial point at the origin.

*Solution.* Set

$$s = t + 1, \quad w = y - 2.$$

Then the initial point  $(t, y) = (-1, 2)$  becomes  $(s, w) = (0, 0)$ , and

$$\frac{dw}{ds} = \frac{dy}{dt}.$$

Since  $y = w + 2$ , the equivalent initial value problem is

$$\boxed{\frac{dw}{ds} = 4 - (w + 2)^3, \quad w(0) = 0}.$$

□

**Problem 2.8.7.** Let  $\phi_0(t) = 0$  and use the method of successive approximations to approximate the solution of

$$y' = 2t^2 + y^2, \quad y(0) = 0.$$

- (a) Calculate  $\phi_1(t), \phi_2(t), \phi_3(t)$ .
- (b) Plot  $\phi_1(t), \phi_2(t), \phi_3(t)$ . Observe whether the iterates appear to be converging.

*Solution.* (a) The successive approximations satisfy

$$\phi_{n+1}(t) = \int_0^t (2s^2 + \phi_n(s)^2) ds.$$

Thus

$$\begin{aligned} \phi_1(t) &= \int_0^t 2s^2 ds = \frac{2}{3}t^3, \\ \phi_2(t) &= \int_0^t \left(2s^2 + \frac{4}{9}s^6\right) ds = \frac{2}{3}t^3 + \frac{4}{63}t^7, \end{aligned}$$

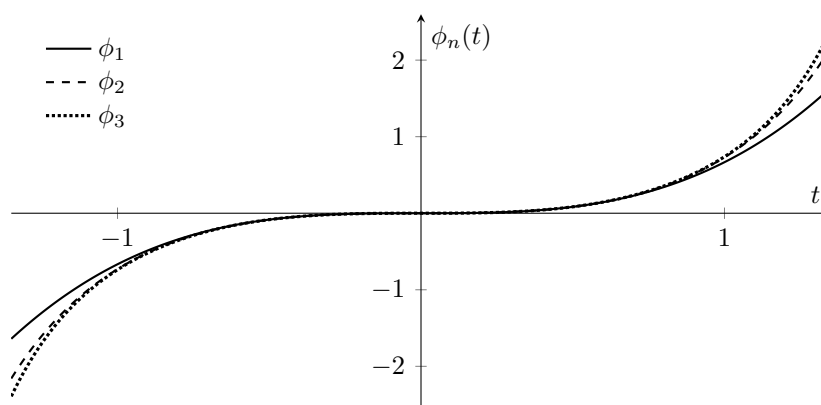
and

$$\begin{aligned} \phi_3(t) &= \int_0^t \left[2s^2 + \left(\frac{2}{3}s^3 + \frac{4}{63}s^7\right)^2\right] ds \\ &= \frac{2}{3}t^3 + \frac{4}{63}t^7 + \frac{16}{2079}t^{11} + \frac{16}{59535}t^{15}. \end{aligned}$$

Hence

$$\boxed{\begin{aligned} \phi_1(t) &= \frac{2}{3}t^3, \\ \phi_2(t) &= \frac{2}{3}t^3 + \frac{4}{63}t^7, \\ \phi_3(t) &= \frac{2}{3}t^3 + \frac{4}{63}t^7 + \frac{16}{2079}t^{11} + \frac{16}{59535}t^{15}. \end{aligned}}$$

(b) The first three iterates are shown below.



Near the initial point, the successive iterates lie increasingly close to one another, which is consistent with convergence of the Picard iterates there.

□

**Problem 3.1.3.** Find the general solution of

$$12y'' - y' - y = 0.$$

*Solution.* The characteristic equation is

$$12r^2 - r - 1 = 0 = (3r - 1)(4r + 1),$$

so

$$r_1 = \frac{1}{3}, \quad r_2 = -\frac{1}{4}.$$

Therefore

$$y = c_1 e^{t/3} + c_2 e^{-t/4}.$$

□

**Problem 3.1.4.** Find the general solution of

$$y'' + 6y' = 0.$$

*Solution.* The characteristic equation is

$$r^2 + 6r = r(r + 6) = 0,$$

with roots  $r_1 = 0$  and  $r_2 = -6$ . Hence

$$y = c_1 + c_2 e^{-6t}.$$

□

**Problem 3.1.9.** Find the solution of the initial value problem

$$y'' + 3y' = 0, \quad y(0) = 0, \quad y'(0) = 3.$$

Sketch the graph of the solution and describe its behavior as  $t$  increases.

*Solution.* The characteristic equation is

$$r(r + 3) = 0,$$

so the general solution is

$$y = c_1 + c_2 e^{-3t}.$$

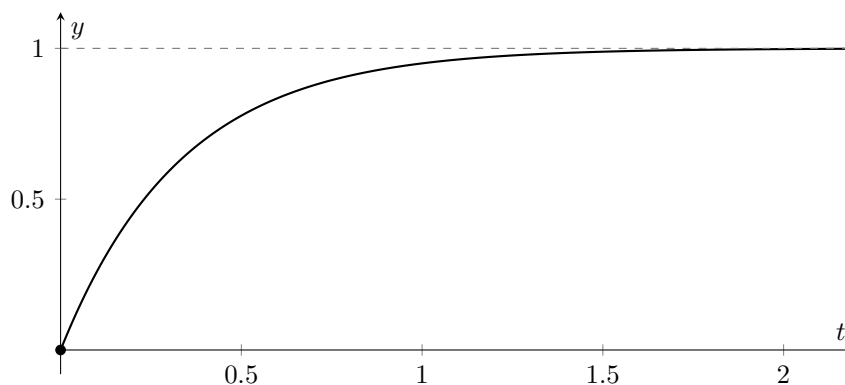
The initial conditions give

$$c_1 + c_2 = 0, \quad -3c_2 = 3,$$

so  $c_1 = 1$  and  $c_2 = -1$ . Therefore

$$\boxed{y = 1 - e^{-3t}}.$$

The graph for  $t \geq 0$  is shown below; the dashed line is the horizontal asymptote  $y = 1$ .



Since  $e^{-3t} \rightarrow 0$  as  $t \rightarrow \infty$ ,

$$\boxed{y(t) \rightarrow 1 \text{ as } t \rightarrow \infty}.$$

The solution increases from 0 and approaches 1 from below.

□

**Problem 3.2.7.** Determine the longest interval in which the initial value problem

$$t(t - 4)y'' + 3ty' + 5y = 2, \quad y(3) = 0, \quad y'(3) = -1$$

is certain to have a unique twice-differentiable solution. Do not attempt to find the solution.

*Solution.* Dividing by  $t(t - 4)$  gives

$$y'' + \frac{3}{t-4}y' + \frac{5}{t(t-4)}y = \frac{2}{t(t-4)}.$$

The coefficients are continuous except at  $t = 0$  and  $t = 4$ . The largest open interval containing the initial point  $t = 3$  on which all coefficients are continuous is

$$\boxed{(0, 4)}.$$

By Theorem 3.2.1, the initial value problem has a unique twice-differentiable solution throughout this interval.

□

**Problem 3.2.18.** Consider the differential equation

$$y'' - y' - 2y = 0.$$

(a) Show that  $y_1(t) = e^{-t}$  and  $y_2(t) = e^{2t}$  form a fundamental set of solutions.

(b) Let

$$y_3(t) = -2e^{2t}, \quad y_4(t) = y_1(t) + 2y_2(t), \quad y_5(t) = 2y_1(t) - 2y_3(t).$$

Are  $y_3(t)$ ,  $y_4(t)$ , and  $y_5(t)$  also solutions of the differential equation?

(c) Determine whether each of the pairs  $\{y_1(t), y_3(t)\}$  and  $\{y_2(t), y_3(t)\}$  forms a fundamental set of solutions.

*Solution.* (a) Direct substitution gives

$$y_1'' - y_1' - 2y_1 = e^{-t} + e^{-t} - 2e^{-t} = 0$$

and

$$y_2'' - y_2' - 2y_2 = 4e^{2t} - 2e^{2t} - 2e^{2t} = 0,$$

so both functions are solutions. Their Wronskian is

$$\begin{aligned} W(y_1, y_2)(t) &= \begin{vmatrix} e^{-t} & e^{2t} \\ -e^{-t} & 2e^{2t} \end{vmatrix} \\ &= 3e^t \neq 0. \end{aligned}$$

Hence  $y_1$  and  $y_2$  form a fundamental set of solutions.

(b) Since  $y_3 = -2y_2$ ,  $y_4 = y_1 + 2y_2$ , and

$$y_5 = 2y_1 - 2y_3 = 2y_1 + 4y_2,$$

each is a linear combination of the solutions  $y_1$  and  $y_2$ . By Theorem 3.2.2, all three are also solutions.

(c) Since  $y_3 = -2y_2$ ,

$$W(y_1, y_3)(t) = -2W(y_1, y_2)(t) = -6e^t \neq 0,$$

so

$$\boxed{\{y_1, y_3\} \text{ is a fundamental set}}.$$

On the other hand,  $y_2$  and  $y_3$  are constant multiples of one another, so their Wronskian is zero. Thus

$$\boxed{\{y_2, y_3\} \text{ is not a fundamental set}}.$$

□

**Problem 3.2.20.** Find the fundamental set of solutions specified by Theorem 3.2.5 for

$$y'' + 5y' + 4y = 0, \quad t_0 = 1.$$

*Solution.* The characteristic equation is

$$r^2 + 5r + 4 = (r + 1)(r + 4) = 0,$$

so it is convenient to write the general solution as

$$y = Ae^{-(t-1)} + Be^{-4(t-1)}.$$

By Theorem 3.2.5,  $y_1$  is the solution satisfying

$$y_1(1) = 1, \quad y_1'(1) = 0.$$

Thus

$$A + B = 1, \quad -A - 4B = 0,$$

which gives  $A = 4/3$  and  $B = -1/3$ . Hence

$$y_1(t) = -\frac{1}{3}e^{-4(t-1)} + \frac{4}{3}e^{-(t-1)}.$$

Similarly,  $y_2$  is the solution satisfying

$$y_2(1) = 0, \quad y_2'(1) = 1.$$

Then

$$A + B = 0, \quad -A - 4B = 1,$$

so  $A = 1/3$  and  $B = -1/3$ . Therefore the specified fundamental set is

$$\begin{cases} y_1(t) = -\frac{1}{3}e^{-4(t-1)} + \frac{4}{3}e^{-(t-1)}, \\ y_2(t) = -\frac{1}{3}e^{-4(t-1)} + \frac{1}{3}e^{-(t-1)}. \end{cases}$$

□

**Problem 3.3.6.** Find the general solution of

$$y'' - 2y' + 8y = 0.$$

*Solution.* The characteristic equation is

$$r^2 - 2r + 8 = 0,$$

with roots

$$r = 1 \pm i\sqrt{7}.$$

Therefore

$$y = c_1 e^t \cos(\sqrt{7}t) + c_2 e^t \sin(\sqrt{7}t).$$

□

**Problem 3.3.13.** Find the solution of the initial value problem

$$y'' - 2y' + 5y = 0, \quad y\left(\frac{\pi}{2}\right) = 0, \quad y'\left(\frac{\pi}{2}\right) = 4.$$

Sketch the graph of the solution and describe its behavior for increasing  $t$ .

*Solution.* The characteristic equation is

$$r^2 - 2r + 5 = 0,$$

with roots  $r = 1 \pm 2i$ . Hence

$$y = e^t(c_1 \cos 2t + c_2 \sin 2t).$$

The first initial condition gives  $c_1 = 0$ . Thus

$$y = c_2 e^t \sin 2t,$$

and

$$y' = c_2 e^t (\sin 2t + 2 \cos 2t).$$

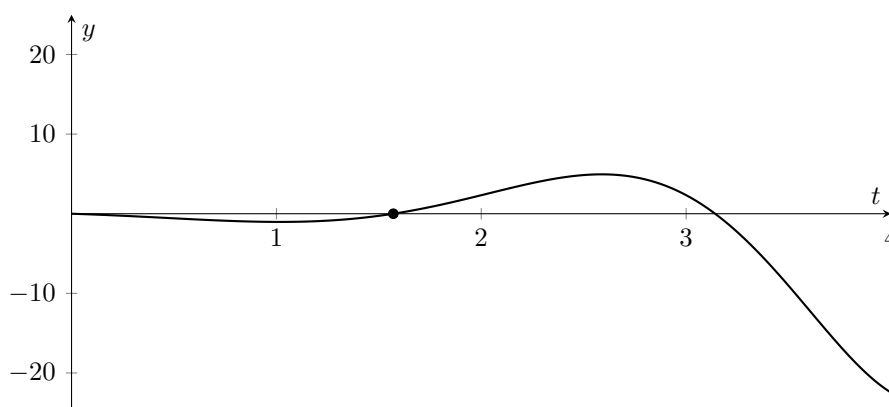
Using  $y'(\pi/2) = 4$  gives

$$-2c_2 e^{\pi/2} = 4, \quad c_2 = -2e^{-\pi/2}.$$

Therefore

$$y = -2e^{t-\pi/2} \sin 2t.$$

The graph is shown below. The marked point is the initial point  $(\pi/2, 0)$ .



The trigonometric factor produces oscillation with period  $\pi$ , while the factor  $e^{t-\pi/2}$  makes the amplitude grow exponentially. Thus the solution is a

growing oscillation.

□

**Problem 3.4.4.** Find the general solution of

$$4y'' - 12y' + 9y = 0.$$

*Solution.* The characteristic equation is

$$4r^2 - 12r + 9 = (2r - 3)^2 = 0,$$

so  $r = 3/2$  is a repeated root. Therefore

$$y = c_1 e^{3t/2} + c_2 t e^{3t/2}.$$

□

**Problem 3.4.19.** Use the method of reduction of order to find a second solution of

$$t^2 y'' + t y' - 4y = 0, \quad t > 0,$$

given that  $y_1(t) = t^2$  is one solution.

*Solution.* Set

$$y = v(t)y_1(t) = v(t)t^2.$$

Then

$$y' = v't^2 + 2vt, \quad y'' = v''t^2 + 4v't + 2v.$$

Substitution into the differential equation gives

$$t^4v'' + 5t^3v' = 0.$$

Since  $t > 0$ , this reduces to

$$tv'' + 5v' = 0.$$

Let  $w = v'$ . Then

$$tw' + 5w = 0,$$

so

$$\frac{w'}{w} = -\frac{5}{t}.$$

Hence

$$w = ct^{-5}, \quad v = -\frac{c}{4}t^{-4} + k.$$

Therefore

$$y = vt^2 = -\frac{c}{4}t^{-2} + kt^2.$$

The term  $kt^2$  is a multiple of the known solution  $y_1$ , while  $t^{-2}$  is not a constant multiple of  $t^2$ . Thus an independent second solution is

$$\boxed{y_2(t) = t^{-2}}.$$

□